

Number Types and Functions to Store Data

1 Number Types

- change of basis

2 Functions to Store Data

- a composite data structure: the list
- an application of shared references

MCS 320 Lecture 7
Introduction to Symbolic Computation
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Binary and Decimal Notations of Numbers

The basic number types are

- the integers, rationals, reals, and complex numbers,
- denoted respectively by $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$.

Internally, all numbers are stored in binary.

The real number 0.1 has no exact binary representation, in contrast to its decimal representation which agrees with $1/10$.

We define

- for any real number,
- given a bound on the size of the denominator,

the nearby rational approximation with a bounded denominator.

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1 Number Types

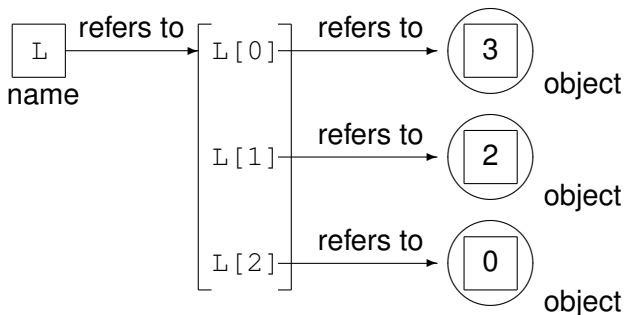
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a composite data structure: the list

After the assignment $L = [3, 2, 0]$, we have:



The $L = [3, 2, 0]$ defined the names $L[0]$, $L[1]$, and $L[2]$.

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Functions to Store Data

Consider the definition of the function:

```
def fun(item, data=[]):  
    data.append(item)  
    print(data)
```

- `data` is the default argument, initialized to a list,
- `data.append(item)` appends `item` to `data`.

The first time the function is called:

- 1 A list is made in memory and assigned to `data`.
- 2 The value of `item` is appended to the list `data`.

The next times the function is called:

- 1 The same list in memory is used as `data`.
- 2 The value of `item` is appended to the list `data`.